Safety systems · Germany

German signals mostly show **what to expect**: a main signal (Hauptsignal) says whether you may pass it and at what speed, a distant signal (Vorsignal) what the next main signal shows. Speeds on signals and boards are **number × 10 km/h**. Four signal families appear on the TSW 7 routes: **H/V** (West), **Ks** (new lines), **Hl** (former DR, e.g. Leipzig – Dresden) and semaphores. **PZB 90** watches every signal, **LZB** takes over on high-speed lines, and the **SiFa** checks that you are awake. Quadrant controls are from TheJAG's German profiles.

Safety systems

PZB 90
Punktförmige Zugbeeinflussung

Track magnets at **1000 Hz** (distant signal), **500 Hz** (about 250 m before the main signal, not at every signal) and **2000 Hz** (main signal). Acknowledge, then brake within the speed curve or PZB brakes.

WHERE Every German route

QUADRANT **ENG 2** Wachsam (acknowledge), **ENG 1** Frei (release), **round aux** Befehl 40; right aux square PZB on / off

LZB
Linienzugbeeinflussung

Continuous cab signalling by cable loop: target speed and distance up to 13 km ahead on the cab display; lineside signals may be dark. Above 160 km/h only under LZB.

WHERE High-speed lines, e.g. Kassel – Würzburg, Nürnberg – Ingolstadt, Köln – Aachen, Leipzig – Dresden

QUADRANT LZB orders and Ende: **ENG 1** (Frei); nothing to press when LZB takes over

SiFa
Sicherheitsfahrschaltung

Driver vigilance: a lamp after about 30 s without a SiFa action, a buzzer 2.5 s later, an emergency brake 2.5 s after that.

WHERE Every German train

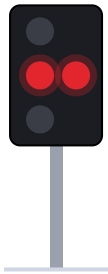
QUADRANT **Gear lever** = SiFa reset; left aux square SiFa on / off

ETCS
European Train Control System

Replacing PZB / LZB on new lines in the real world.

WHERE Not covered here; TSG's BR 101 Expert has an ETCS variant

H/V light signals (Hauptsignal / Vorsignal)



Hp 0
Stop (Halt). One or two red lights.

DRIVE Stop before the signal; 2000 Hz magnet is live.



Hp 1
Proceed (Fahrt) at line speed.

DRIVE Drive on at the timetable / board speed.



Hp 2 (+ Zs 3)
Proceed slowly: 40 km/h, or the Zs 3 number × 10.

DRIVE Be at the speed at the signal; applies through the points area beyond.



Vr 0
Expect stop at the next main signal.

DRIVE Wachsam at the 1000 Hz magnet, then brake to stop.



Vr 1
Expect proceed at the next main signal.

DRIVE Keep going.



Vr 2 (+ Zs 3v)
Expect slow: 40 km/h, or the Zs 3v number × 10.

DRIVE Wachsam, brake to the speed by the main signal.

Ks signals (Kombinationssignal)

One head is main and distant signal at once. A white Zs 3 number above = speed from this signal; a yellow Zs 3v number below = speed at the next signal.



Ks 1
Proceed; next signal shows proceed.

DRIVE Drive on at line speed.



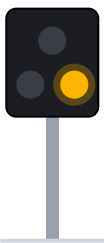
Ks 1 + Zs 3
Proceed at max 80 km/h from this signal.

DRIVE Be at 80 at the signal.



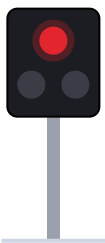
Ks 1 flashing + Zs 3v
Proceed; next signal limits to 60 km/h.

DRIVE Wachsam at 1000 Hz (live for Zs 3v 1–6), brake to 60 by the next signal.



Ks 2
Expect stop at the next signal.

DRIVE Wachsam at 1000 Hz, brake to stop.



Hp 0
Stop.

DRIVE Stop before the signal.

Sources: Eisenbahn-Signalordnung (ESO) · DB Ril 301 Signalbuch · DB Ril 483 (PZB 90, LZB) · de.wikipedia: PZB, LZB, Sicherheitsfahrschaltung, Signale H/V, Ks, Hl

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Hl signals (former Deutsche Reichsbahn)

Upper lamps = the next signal, lower yellow = this signal (40 km/h; with a yellow light strip below 60; a green strip alone = 100).



Hl 1
Line speed; next signal proceed.

DRIVE Drive on.



Hl 7
Line speed; expect 40 (or 60) at the next signal.

DRIVE Wachsam, brake to 40.



Hl 10
Line speed; expect stop at the next signal.

DRIVE Wachsam, brake to stop.



Hl 3a
40 km/h here; next signal proceed.

DRIVE Be at 40 at the signal.



Hl 12a
40 km/h here; expect stop at the next signal.

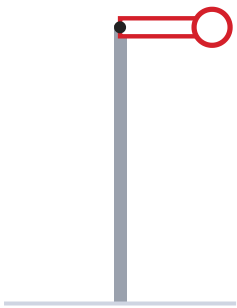
DRIVE Be at 40, then brake to stop.



Hp 0
Stop.

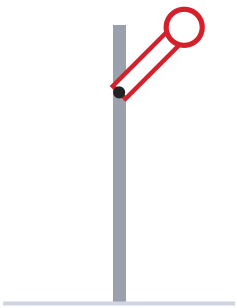
DRIVE Stop before the signal.

Semaphores (Formsignale)



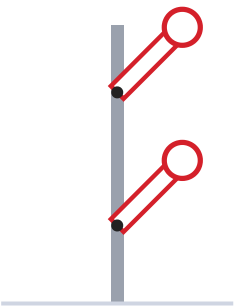
Hp 0
Arm horizontal: stop.

DRIVE Stop before the signal.



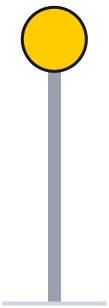
Hp 1
Arm raised: proceed.

DRIVE Drive on.



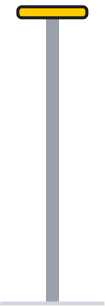
Hp 2
Two arms raised: proceed at 40 km/h.

DRIVE Be at 40 at the signal.



Vr 0
Yellow disc face-on: expect stop.

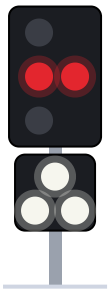
DRIVE Wachsam, brake to stop.



Vr 1
Disc turned flat: expect proceed.

DRIVE Keep going.

Additional signals and boards



Zs 1
Substitute signal: pass the red signal without a written order. Ks / Hl: one flashing white light.

DRIVE Hold Befehl 40 over the 2000 Hz magnet, max 40.



Zs 7
Caution: pass the red signal on sight.

DRIVE Befehl 40, drive on sight, max 40.



Sh 1
Shunting allowed (dwarf or on a main signal).

DRIVE Shunt on sight; not a train movement.



Lf 6
Speed restriction ahead: 80 km/h from the Lf 7.

DRIVE Brake to reach 80 at the Lf 7.



Lf 7
Speed limit 80 km/h from here.

DRIVE Lower: from the front. Higher: once the rear has passed.



Ne 2
Distant signal board: a distant signal stands here.

DRIVE Alone (no distant signal): 1000 Hz always live; Wachsam, expect stop. A dark distant signal here = Vr 0.

Sources: Eisenbahn-Signalordnung (ESO) · DB Ril 301 Signalbuch · DB Ril 483 (PZB 90, LZB) · de.wikipedia: PZB, LZB, Sicherheitsfahrschaltung, Signale H/V, Ks, Hl

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Safety systems · Germany

PZB 90 magnets and speed checks

Magnet	Where	Acknowledge	Train type 0 (85 lamp)	M (70)	U (55)
1000 Hz	At the distant signal showing caution	Wachsam within **4 s**	165 → 85 km/h in 23 s	125 → 70 in 29 s	105 → 55 in 38 s
500 Hz	About 250 m before some main signals at stop	–	65 → 45 km/h over 153 m	50 → 35	40 → 25
2000 Hz	At the main signal at stop	–	Immediate emergency brake		
Restrictive	Below 10 km/h for 15 s, or a stop, inside a 1000 Hz (45) or 500 Hz (25) check	–	45 km/h (1000 Hz), 25 km/h (500 Hz)		

How to drive under PZB 90, LZB and SiFa

1000 Hz: distant signal at caution

1. Passing a Vr 0 / Vr 2 / Ks 2 / Ks 1 flashing with Zs 3v ≤ 6 / Hl 7 / Hl 10 the PZB sounds: press **Wachsam** (ENG 2) within 4 s.
2. The 1000 Hz lamp lights for 1250 m: be below the train-type speed within the time.
3. Brake to the signal's speed or to a stop.
4. If the signal clears, press **Frei** (ENG 1) after 700 m to end the check.

500 Hz and 2000 Hz: approaching red

1. 250 m before a red main signal the 500 Hz magnet checks a falling curve to 45 / 35 / 25 km/h.
2. Stop before the signal: the 2000 Hz magnet brakes at once if you pass it at stop.
3. Frei cannot end the 500 Hz check.

Befehl 40: passing a red signal

1. Only with Zs 1, Zs 7 or a written order.
2. Hold **Befehl 40** (round aux) while the front passes the 2000 Hz magnet.
3. Max 40 km/h to the next main signal; on sight only with Zs 7 or when ordered.

Emergency brake (Zwangsbremsung)

1. The train stops; the flashing lamp shows why: yellow 1000 Hz, red 500 Hz, white Befehl 40.
2. At a stand, press **Frei** (ENG 1).
3. Release the brake and drive on, restrictive (45 / 25 km/h) where the check is still running.

Start and restrictive mode

- After switching on or a stop inside a check, PZB runs restrictive: the **70 and 85 lamps flash alternately**.
- Start programme: max 45 km/h for 550 m; Frei ends it when no check is running.
- Pick the train type in the cab (0 passenger, M / U freight).

LZB

- Entering: **Ü** and **B** light; nothing to press.
- Follow **V-Soll** on the display; signals may be dark. Leave the AFB (flap lever, where fitted) at line speed: LZB lowers the set-point itself.
- **Ende** flashes about 1700 m before the exit: press **Frei** (ENG 1) within 10 s. At the exit Ü goes out and PZB rules again.
- Ü flashing (transmission lost) or **V 40**: Frei, then max 40 (V 40 on sight). **E 40**: no button, max 40.

SiFa

1. Keep the SiFa on (left aux square).
2. When the SiFa lamp lights, flick the **gear lever**; any SiFa action resets the timer.
3. Missed: emergency brake; stop, reset the SiFa, release and drive on.

Before departure: SiFa and PZB on: aux squares lit on most locos, unlit where the square is the Störschalter or isolation switch (ICE, S-Bahn 42x, 612); see the train's manual. Set the right train type, check the signal. With PZB restrictive after start-up keep below 45 km/h until the check ends.

Sources: Eisenbahn-Signalordnung (ESO) · DB Ril 301 Signalbuch · DB Ril 483 (PZB 90, LZB) · de.wikipedia: PZB, LZB, Sicherheitsfahrschaltung, Signale H/V, Ks, Hl

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